

Letters of Euler
ON NATURAL PHILOSOPHY

Transcribed by : Uncle Jeffy
for a Canadian Princess and Prince

Dearest Juliet and Jackson,

First Class of known truths. Conviction ^{that} ~~and~~ things exist externally, corresponding to the ideas represented by the senses. Objection of the Pyrrhonists. Reply.

We include in the first class of known truths those which we acquire immediately by means of the senses. I have already remarked, that they not only supply the soul with certain representations relative to the changes produced in a part of the brain; but that they excite there a conviction of the real existence of things external, corresponding to the ideas which the senses present to us.

The soul is frequently compared to a man shut up in a dark room, in which the images of external objects are represented on the wall by means of a glass. This comparison is tolerably just, as far as it respects the man looking at the images on the wall; for this act is sufficiently similar to that of the soul, contemplating the impressions made in the brain; but the comparison appears to me extremely defective, as far as it respects the conviction that the objects

which occasion these images really exist.

The man in the dark room will immediately suspect the existence of these objects; and if he has no doubt about the matter, it is because he has been out of doors, and has seen them; besides this, knowing the nature of his glass, he is assured that nothing can be represented on the wall but the images of the objects which are without the chamber before the glass. But this is not the case with the soul; it has never quitted its place of residence to contemplate the objects themselves; and it knows still less the construction of the sensitive organs, and the nerves which terminate in the brain. It is nevertheless much more powerfully convinced of the real existence of objects than our man in the dark room possibly can be.

I am apprehensive of no objection on this subject, the thing being too clear of itself to admit any, though we do not know the true foundation of it. No one ever entertained any ~~doubt~~ doubt about it, except certain visionaries who have bewildered themselves in their own reveries. Though they say that they doubt the existence of external objects, they entertain no such doubt in fact; for why would they have affirmed it, unless they had believed the existence of other men, to whom they wished to communicate their extravagant opinions.

This conviction respecting the existence of the things whose images the senses represent, appears not only in men of every age and condition, but likewise in all animals.

The dog which barks at me has no doubt of my existence, but his soul perceives but a slight image of my person. Hence I conclude, that this conviction is essentially connected with our sensations, and that the truths which the senses convey to us are ~~as~~ as well founded as the most undoubted truths of geometry.

Without this conviction no human society could subsist, for we should be continually falling into the greatest absurdities, and the ~~grossest~~ grossest contradictions.

Were the peasantry to dream of doubting about the existence of their bailiff, or soldiers about that of their officers, into what confusion should ~~be~~ be plunged! Such absurdities are entertained only by philosophers; any other giving himself up to them would be considered as having lost his reason. Let us then acknowledge this conviction as one of the principal laws of nature, and that it is complete, though we are absolutely ignorant of its true reasons, and very far from being able to explain them in an intelligible manner.

However important this reflection may be, it is by no means, however, exempted from difficulties; but were they

ever so great, and though it might be impossible for us to solve them, they do not in the smallest degree affect the truth I have just established, and which we ought to consider as the most solid foundation of human knowledge.

It must be allowed that our senses sometimes deceive us; and ~~therefore~~ hence it is that those ~~the~~ subtle philosophers who value themselves on doubting of every thing deduce the consequence, that we ought never to depend on our senses. I have perhaps oftener than once met an unknown person in the street, whom I mistook for an acquaintance: as I was deceived in that instance, nothing prevents my being always deceived; and I am, therefore, never assured that the person to whom I speak is in reality the one I imagine.

Were I to go to Magdeburg, and to present myself to your highness, I ought always to be apprehensive of grossly mistaking: nay, perhaps I should not be at Magdeburg, for there are instances of a man's sometimes taking one city for another. It is even possible I may never have had the happiness of seeing you, but was always under the power of delusion when I thought myself to be enjoying that felicity.

Such are the natural consequences resulting from sentiments of certain philosophers; and you must be abundantly sensible that they not only lead to manifest absurdity, but have also a tendency to dissolve all the bonds of society.

7th April, 1761.

Notes from Uncle Jeffy :

Euler has just introduced us to the

ALLEGORY OF THE CAVE

This is an old model for how we only can get an ~~image~~ image of reality, not the whole movie. Like how the # of hours of video uploaded to Youtube per day is 720,000 hours!

It took your uncle Jeffy a while to realize that the sun is going to rise again, tomorrow. It is good to get to bed on time. Your toys will be ready for you to play with them again tomorrow.

Euler said something naughty about a dog having less of a soul than humans. We now know this is not true; but, a dog is not as important as a human baby. He just had an old man way of saying this.

Euler is laying a foundation like in the way you first strengthen your legs so you can walk to the park, school, or gym to strengthen other parts of you.

this is plenty abstract for our purposes...
next we are getting into Colors! and Sound!

Love,

Jefferson